

B.1 Assume the following likelihoods for each word being part of a positive or negative movie review, and equal prior probabilities for each class.

	pos	neg
I	0.09	0.16
always	0.07	0.06
like	0.29	0.06
foreign	0.04	0.15
films	0.08	0.11

What class will Naive bayes assign to the sentence “I always like foreign films.”?

$$P(\text{“I always like foreign films”} \mid \text{pos}) = 0.09 \cdot 0.07 \cdot 0.29 \cdot 0.04 \cdot 0.08$$

- $0.09 \times 0.07 = 0.0063$
- $0.0063 \times 0.29 = 0.001827$
- $0.001827 \times 0.04 = 0.00007308$
- $0.00007308 \times 0.08 = \mathbf{0.0000058464}$

$$\Rightarrow P(\text{sentence} \mid \text{pos}) \approx 5.8464 \times 10^{-6}$$

$$P(\text{sentence} \mid \text{neg}) = 0.16 \cdot 0.06 \cdot 0.06 \cdot 0.15 \cdot 0.11$$

- $0.16 \times 0.06 = 0.0096$
- $0.0096 \times 0.06 = 0.000576$
- $0.000576 \times 0.15 = 0.0000864$
- $0.0000864 \times 0.11 = \mathbf{0.000009504}$

$$\Rightarrow P(\text{sentence} \mid \text{neg}) \approx 9.504 \times 10^{-6}$$

$P(\text{sentence} \mid \text{neg}) > P(\text{sentence} \mid \text{pos}) \Rightarrow$ NB predicts the sentence is **NEGATIVE.**